LITERATURE SURVEY
BERT VS GPT VS T5 – MODEL SELECTION FOR HR AUTOMATION

Separate Literature Survey Document

Course: Generative AI and AI Platform Engineering (25CS2207E)

Academic Year: 2026–2027 | Section 8 | Team 23

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1. Introduction

Literature survey is the systematic study of previously published research related to a project. It helps identify the methods already used, the findings reported by researchers, the limitations of existing approaches, and the research gap that the proposed project intends to address.

The present project, "BERT vs GPT vs T5 – Model Selection for HR Automation," examines how three transformer model families can support different Human Resource Management (HRM) activities. The project considers BERT for resume classification and candidate-job matching, GPT for an HR question-answering assistant, and T5 for resume or HR-document summarization. This survey therefore covers (i) foundational transformer papers, (ii) research on AI in recruitment and HRM, and (iii) research on ethical and responsible use of AI in employment-related processes.

The survey is organized thematically rather than treating the models as interchangeable. BERT, GPT, and T5 were developed with different learning objectives and output styles; the most appropriate evaluation method depends on the HR task being performed.

2. Objectives of the Literature Survey

- To understand the foundations and capabilities of BERT, GPT, and T5.
- To review existing research on AI-supported recruitment, resume processing, and HR assistance.
- To identify how transformer-based methods are used in language understanding and text generation.
- To examine ethical concerns such as bias, fairness, privacy, transparency, and human accountability.
- To identify a research gap and establish the need for a task-oriented comparison of the three model families.

3. Review of Related Literature

3.1 Foundational Transformer Models

3.1.1 BERT – Bidirectional Encoder Representations from Transformers

Devlin, Chang, Lee, and Toutanova (2018) introduced BERT, a language representation model that uses bidirectional contextual learning. Unlike approaches that process context only from left to right or right to left, BERT learns representations using both directions of context. The paper established a pre-training and fine-tuning approach: a model is first trained on large text collections and is then adapted to a particular downstream task.

For this project, BERT is relevant to resume classification and candidate-job matching. A BERT-based classifier can be fine-tuned to assign resumes to predefined job-related categories, while sentence-embedding variants can be used to compare the semantic similarity of resume content and job descriptions. The original BERT paper does not specifically study recruitment or prove that BERT is fair or accurate for hiring. The project must therefore validate its chosen BERT checkpoint using a clearly labelled HR dataset.

Reference: Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805. https://arxiv.org/abs/1810.04805

3.1.2 GPT – Generative Pre-trained Transformer

Radford, Narasimhan, Salimans, and Sutskever (2018) described generative pre-training for language understanding. The approach trains a transformer language model to predict text and then adapts the learned representations to downstream tasks. This work is an early foundation for the GPT model family and the use of large-scale generative pre-training.

GPT-style models are relevant to the HR chatbot because they can generate natural-language answers to employee questions. In an HR application, however, fluent language does not guarantee factual correctness. The chatbot should use approved HR policy documents as its knowledge source, provide references where possible, and clearly indicate when the answer cannot be found. The foundational GPT paper is not an HR chatbot evaluation; it provides the model-family background only.

Reference: Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). Improving language understanding by generative pre-training. OpenAI. https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

3.1.3 T5 – Text-to-Text Transfer Transformer

Raffel and colleagues (2020) introduced the Text-to-Text Transfer Transformer (T5). The central idea is to represent many NLP problems in a unified text-to-text format: the model receives text and generates text as output. This makes the framework adaptable to tasks such as translation, summarization, and question answering.

T5 is relevant to the proposed resume and HR-document summarization module. A summarization model can convert a long document into a shorter text containing key qualifications, skills, experience, or policy points. The generated summary must be checked against the original document because a model can omit important facts or produce unsupported statements. The T5 paper establishes the general framework, not its measured performance on the team's own resume dataset.

Reference: Raffel, C., Shazeer, N., Roberts, A., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. Journal of Machine Learning Research, 21(140), 1–67. https://jmlr.org/papers/v21/20-074.html

3.2 AI in Human Resource Management and Recruitment

3.2.1 AI-Based Human Resource Management Tools and Techniques: A Systematic Literature Review (2023)

The 2023 systematic review published in Procedia Computer Science examines AI-based tools and techniques used in HRM. It discusses AI applications in recruitment, performance management, workforce management, and employee engagement. The review describes how AI can support activities such as candidate screening and other HR processes, while also recognizing that conventional HR activities can be time-intensive and may involve bias.

Relevance to the project: this work establishes the wider context for applying AI to HR operations. It supports the motivation for automating text-heavy activities, but it does not provide a direct experimental comparison of BERT, GPT, and T5 on the same HR dataset.

Reference: AI-Based Human Resource Management Tools and Techniques; A Systematic Literature Review. (2023). Procedia Computer Science, 229, 367–377. https://doi.org/10.1016/j.procs.2023.12.039

3.2.2 Role of Artificial Intelligence in Employee Recruitment: Systematic Review and Future Research Directions (2025)

Dadaboyev and colleagues (2025) reviewed 49 peer-reviewed articles indexed in the Web of Science and published between 2018 and 2025. The review discusses AI-supported recruitment activities such as automated resume parsing, chatbot-led initial screening, predictive analytics for job fit, and AI-assisted video interview analysis. It also identifies ethical concerns, including algorithmic bias arising from training data and the need for transparency.

Relevance to the project: the review directly supports the resume-screening and HR-chatbot use cases. It also shows why evaluation should not be limited to speed: fairness, transparency, and responsible use are important. Since the paper is a review of AI recruitment research broadly, it should not be interpreted as evidence that one of BERT, GPT, or T5 is best.

Reference: Dadaboyev, S. M. U., Abdullayeva, J., Abbosova, N., Suleymenova, A., et al. (2025). Role of artificial intelligence in employee recruitment: systematic review and future research directions. Discover Artificial Intelligence, 3, Article 99. https://link.springer.com/article/10.1007/s44282-025-00246-w

3.2.3 AI-Driven Research in the Recruitment and Selection Process: Application of an AI Taxonomy With a Systematic Literature Review (2025)

Herold and Roedenbeck (2025) conducted a systematic literature review of AI in the recruitment and selection process, supported by a concept matrix and computational literature review using NLP. Their study aimed to identify which recruitment stages receive research attention and which AI algorithms are used. The paper reports that assessment and selection, as well as processing incoming applications, are important areas of AI research. It also discusses the use of NLP and machine-learning algorithms, including BERT-related approaches, in the literature.

Relevance to the project: this work provides support for studying BERT in resume processing and candidate assessment. It also demonstrates the value of organizing AI techniques according to the stage of the recruitment process. The review maps research; it is not a controlled benchmark comparing BERT, GPT, and T5.

Reference: Herold, M., & Roedenbeck, M. R. H. (2025). AI-driven research in the recruitment and selection process: Application of an AI taxonomy with a systematic literature review. SAGE Open. https://doi.org/10.1177/21582440251361746

3.2.4 How to Use Generative AI as a Human Resource Management Assistant (2024)

Aguinis, Beltran, and Cope (2024) discuss how generative AI can be used as an assistant in human resource management. The paper is relevant to the idea of using a generative model to support HR professionals with information and communication tasks. It provides an HR-focused perspective for considering generative AI as an assistant rather than simply as a general text-generation system.

Relevance to the project: it supports the motivation for an HR question-answering assistant. In implementation, the chatbot should be restricted to approved organizational information and should not make unsupported claims about employment conditions or company policy. The article is an HRM-focused discussion and should not be presented as a controlled performance test of the team's GPT implementation.

Reference: Aguinis, H., Beltran, J. R., & Cope, A. (2024). How to use generative AI as a human resource management assistant. Organizational Dynamics, 53(1), 101029. https://doi.org/10.1016/j.orgdyn.2023.101029

3.2.5 Large Language Models in Human Resource Management: A Systematic Literature Review of Applications, Open Issues and Future Research Directions (2025/2026)

Dasaklis and colleagues prepared a systematic review of LLM-enabled HRM. The review classifies the technical footprint of models into discriminative and generative approaches and maps them to HR functions

such as recruitment, training, and performance evaluation. It identifies recurring issues involving robustness, bias, explainability, transparency, and governance, and emphasizes layered human oversight and auditability.

A later journal listing in Technological Forecasting and Social Change (October 2026) describes a review synthesizing 141 studies. A 2025 SSRN version of the work is also available. The versions should be checked before final citation because the manuscript and final journal publication may differ. The review is particularly useful for framing the distinction between models used to understand/classify text and models used to generate text.

Reference: Dasaklis, T. K., Giannopoulos, P. G., Koutras, D., Malamas, V., & Chountalas, P. T. (2025; journal publication listed 2026). Large language models in human resource management: A systematic literature review of applications, open issues and future research directions. Technological Forecasting and Social Change, 231, 124800. https://doi.org/10.1016/j.techfore.2026.124800

3.2.6 Application of LLM Agents in Recruitment: A Novel Framework for Automated Resume Screening (2024)

Gan, Zhang, and Mori (2024) proposed a large-language-model agent framework for automating resume screening. The framework converts unstructured resume text into structured information through sentence classification, then summarizes and grades resumes to support the recruitment decision process. The authors constructed a dataset from actual resumes and simulated a screening workflow. They report that the automated framework was approximately 11 times faster than traditional manual screening. Fine-tuning improved the F1-score to 87.73% in the resume sentence-classification phase, and the fine-tuned model outperformed the GPT-3.5 baseline in resume summarization and grading.

Relevance to the project: this is a directly related recruitment study because it covers resume classification, summarization, grading, and decision support. It provides useful task-level evaluation ideas, including F1-score for classification and text-generation measures for summarization. Its framework can inform the proposed BERT-based resume processing and T5-based summarization workflows. However, the paper evaluates its own agent framework and selected language-model configurations; it should not be described as a controlled, same-dataset comparison of the team's exact BERT, GPT, and T5 implementations.

Reference: Gan, C., Zhang, Q., & Mori, T. (2024). Application of LLM agents in recruitment: A novel framework for automated resume screening. Journal of Information Processing, 32, 881–893. https://doi.org/10.2197/ipsjjip.32.881

3.2.7 Towards Optimizing and Evaluating a Retrieval Augmented QA Chatbot using LLMs with Human-in-the-Loop (2024)

Afzal, Kowsik, Fani, and Matthes (2024) describe the development and evaluation of an HR support chatbot with SAP SE. The system uses Retrieval-Augmented Generation (RAG) to retrieve relevant information before generating answers to employee questions. Domain experts participate in dataset collection, prompt optimization, and evaluation. The study benchmarks OpenAI models and uses BERT and LongT5 as baselines. Its findings report that GPT-4 performs better than the other evaluated models in the human assessment, while BERT-based retrieval provides a useful baseline for finding relevant HR information.

Relevance to the project: this paper directly supports the GPT-based HR question-answering module and demonstrates how BERT can support retrieval in an HR knowledge system. It also highlights the importance of grounding chatbot answers in approved HR documents and involving human experts in evaluation. The paper focuses on HR question answering and RAG; it does not establish that GPT is universally best for all HR tasks or directly compare the team's complete BERT/GPT/T5 workflows.

Reference: Afzal, A., Kowsik, A., Fani, R., & Matthes, F. (2024). Towards optimizing and evaluating a retrieval augmented QA chatbot using LLMs with human-in-the-loop. Proceedings of the Fifth Workshop on Data Science with Human-in-the-Loop (Language Advances), 4–16. https://aclanthology.org/2024.dash-1.2/

3.3 Fairness, Ethics, and Responsible AI in Recruitment

3.3.1 Ethics of AI-Enabled Recruiting and Selection: A Review and Research Agenda (2022)

Hunkenschroer and Luetge (2022) review ethical issues related to AI-enabled recruiting and selection. The work is important because hiring is a high-impact context: automated systems can influence who receives attention or advances through a recruitment process. Ethical analysis therefore needs to consider fairness, transparency, accountability, and the role of human decision-makers.

Relevance to the project: the resume screening module should be used only as decision support. A recruiter should review the evidence behind model outputs, and the project should avoid treating a similarity score as an objective measure of a candidate's overall suitability.

Reference: Hunkenschroer, A. L., & Luetge, C. (2022). Ethics of AI-enabled recruiting and selection: A review and research agenda. Journal of Business Ethics, 178, 977–1007. https://doi.org/10.1007/s10551-022-05049-6

3.3.2 Mapping the Use of Large Language Models in Hiring Decisions: A Scoping Review (2026)

This scoping review maps academic literature on the use of LLMs in hiring across recruitment stages. It reports that the literature is expanding but uneven, with substantial attention to early hiring stages and efficiency/performance outcomes. It also notes that ethical risks are widely recognized while empirical validation of mitigation strategies remains limited.

Relevance to the project: the work reinforces the need to evaluate more than speed and to document whether fairness controls are actually tested. It also supports a cautious framing of LLMs as tools whose effects depend on the context and governance arrangements.

Reference: Mapping the use of large language models in hiring decisions: A scoping review. (2026). PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC13021839/

3.3.3 AI and Generative AI for Recruitment and Talent Management: A PRISMA-Based Systematic Literature Review (2026)

Saouabe and colleagues (2026) present a PRISMA-based systematic review of 21 studies published between January 2025 and May 2026. The review covers AI-driven recruitment, resume information extraction,

semantic matching, workforce forecasting, generative AI, and responsible AI. Its abstract describes a shift from keyword-based applicant tracking toward transformer-based models, LLM-based systems, retrieval-augmented generation, and agentic AI. It also identifies unresolved issues including bias, explainability, privacy, hallucination, and a lack of standardized benchmarks.

Relevance to the project: this recent review closely matches the project's combination of resume understanding and generative HR assistance. It also highlights the need for clear, task-specific benchmarks. Because the publication is very recent, the final submitted version should verify the volume, issue, and page details from the journal record.

Reference: Saouabe, A., Banoui, D., Tizra, K., Mazar, M., & Mourtaji, I. (2026). AI and generative AI for recruitment and talent management: A PRISMA-based systematic literature review. International Journal of Advanced Computer Science and Applications, 17(10). https://doi.org/10.14569/IJACSA.2026.0171008

4. Comparative Synthesis of the Literature

The reviewed studies can be grouped into three connected areas. First, the foundational model papers explain the technical differences among BERT, GPT, and T5. Second, HRM and recruitment reviews show that AI is being considered for resume processing, candidate assessment, employee communication, and other HR activities. Third, ethics and governance research warns that hiring systems can create or reproduce unfair outcomes if they are deployed without appropriate oversight.

Study / year	Main focus	Contribution to this project	Limitation for the present project
Devlin et al. (2018)	BERT architecture and pre-training	Foundation for contextual classification and text understanding	Not an HR recruitment experiment
Radford et al. (2018)	Generative pre-training / GPT	Foundation for generative language and chatbot responses	Not an HR chatbot benchmark
Raffel et al. (2020)	T5 text-to-text framework	Foundation for summarization and text generation	Not a resume-specific evaluation
AI-Based HRM review (2023)	AI tools across HRM	Broad motivation for AI-supported HR tasks	Does not compare the three model families
Aguinis et al. (2024)	Generative AI as HR assistant	HR context for an employee query assistant	Not a controlled comparison of GPT models
Dadaboyev et al. (2025)	AI in employee recruitment	Resume parsing, initial screening, job-fit and ethics context	Broad AI review rather than BERT/GPT/T5 benchmark

Herold & Roedenbeck (2025)	AI in recruitment stages and algorithms	Supports recruitment-stage mapping and BERT/NLP relevance	Maps literature; does not test the team's models
Dasaklis et al. (2025/2026)	LLMs in HRM	Discriminative/generative model framing and governance	Review-level synthesis, not the team's task dataset
Saouabe et al. (2026)	AI and GenAI recruitment review	Recent coverage of semantic matching, LLMs, RAG and responsible AI	Reviews selected recent studies; not a direct model benchmark
Gan et al. (2024)	LLM-agent framework for resume screening	Resume classification, summarization, grading, and recruitment workflow	Not a unified controlled comparison of the team's exact BERT/GPT/T5 workflows
Afzal et al. (2024)	RAG-based HR question-answering chatbot	GPT-based HR assistance, BERT retrieval baseline, human-in-the-loop evaluation	Focuses on HR Q&A rather than resume screening and summarization

5. Research Gap

The literature provides strong foundations for the individual model families and documents a growing range of AI applications in HRM. However, the reviewed papers do not provide a single controlled, student-scale experiment that evaluates the team's three proposed workflows—BERT-based resume classification/matching, GPT-based HR question answering, and T5-based summarization—using a common implementation environment and clearly reported task-specific metrics.

The gap is therefore not that BERT, GPT, or T5 have never been studied. Rather, the proposed project addresses the practical question of how to select a suitable model family for different HR tasks, while documenting quality, response time, resource needs, and responsible-use considerations. Because the tasks are different, the project should not compare raw accuracy across all three models as if they were solving the same problem. Each workflow needs its own suitable test data and evaluation measures.

6. How the Present Project Addresses the Gap

- BERT workflow: evaluate resume classification using accuracy, precision, recall, F1-score, and a confusion matrix; if semantic matching is used, include human assessment of relevance.

- GPT workflow: test HR questions against approved policy/FAQ content and evaluate correctness, relevance, groundedness, completeness, and appropriate abstention when information is unavailable.
- T5 workflow: assess summaries for factual consistency, coverage of important information, readability, and conciseness; use ROUGE only if suitable reference summaries are prepared.
- System-level comparison: report response time, model/checkpoint identity, hardware or API configuration, and resource requirements.
- Responsible-use checks: keep a human reviewer in the loop, avoid protected-attribute inference, protect personal data, and record model limitations.

7. Conclusion

The literature survey shows that BERT, GPT, and T5 provide different capabilities that can be mapped to different HR automation tasks. BERT provides contextual representations that can support classification and matching. GPT-style models generate conversational responses and can support an HR assistant when answers are grounded in trusted policy information. T5 provides a unified text-to-text framework that can support summarization. Research on AI in HRM and recruitment identifies potential efficiency benefits, but also emphasizes bias, privacy, transparency, explainability, and human accountability. Based on these findings, the proposed project will implement and evaluate each model in a task-appropriate workflow and will recommend model suitability based on measured evidence rather than assuming one model is best for every HR task.

8. References

1. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805. https://arxiv.org/abs/1810.04805
2. Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). Improving language understanding by generative pre-training. OpenAI. https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf
3. Raffel, C., Shazeer, N., Roberts, A., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. Journal of Machine Learning Research, 21(140), 1–67. https://jmlr.org/papers/v21/20-074.html
4. AI-Based Human Resource Management Tools and Techniques; A Systematic Literature Review. (2023). Procedia Computer Science, 229, 367–377. https://doi.org/10.1016/j.procs.2023.12.039
5. Aguinis, H., Beltran, J. R., & Cope, A. (2024). How to use generative AI as a human resource management assistant. Organizational Dynamics, 53(1), 101029. https://doi.org/10.1016/j.orgdyn.2023.101029
6. Dadaboyev, S. M. U., Abdullayeva, J., Abbosova, N., Suleymenova, A., et al. (2025). Role of artificial intelligence in employee recruitment: Systematic review and future research directions. Discover Artificial Intelligence, 3, Article 99. https://link.springer.com/article/10.1007/s44282-025-00246-w

7. Herold, M., & Roedenbeck, M. R. H. (2025). AI-driven research in the recruitment and selection process: Application of an AI taxonomy with a systematic literature review. SAGE Open. https://doi.org/10.1177/21582440251361746
8. Hunkenschroer, A. L., & Luetge, C. (2022). Ethics of AI-enabled recruiting and selection: A review and research agenda. Journal of Business Ethics, 178, 977–1007. https://doi.org/10.1007/s10551-022-05049-6
9. Dasaklis, T. K., Giannopoulos, P. G., Koutras, D., Malamas, V., & Chountalas, P. T. (2025; journal publication listed 2026). Large language models in human resource management: A systematic literature review of applications, open issues and future research directions. Technological Forecasting and Social Change, 231, 124800. https://doi.org/10.1016/j.techfore.2026.124800
10. Mapping the use of large language models in hiring decisions: A scoping review. (2026). PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC13021839/
11. Saouabe, A., Banoui, D., Tizra, K., Mazar, M., & Mourtaji, I. (2026). AI and generative AI for recruitment and talent management: A PRISMA-based systematic literature review. International Journal of Advanced Computer Science and Applications, 17(10). https://doi.org/10.14569/IJACSA.2026.0171008

Gan, C., Zhang, Q., & Mori, T. (2024). Application of LLM agents in recruitment: A novel framework for automated resume screening. Journal of Information Processing, 32, 881–893. https://doi.org/10.2197/ipsjjip.32.881

Afzal, A., Kowsik, A., Fani, R., & Matthes, F. (2024). Towards optimizing and evaluating a retrieval augmented QA chatbot using LLMs with human-in-the-loop. Proceedings of the Fifth Workshop on Data Science with Human-in-the-Loop (Language Advances), 4–16. https://aclanthology.org/2024.dash-1.2/

9. Notes for Final Submission

- Verify the final bibliographic details and access the full text of every paper before submitting the survey.
- Use your institution's required referencing format (APA, IEEE, or another prescribed style) consistently.
- Do not describe a review paper as an experiment, and do not claim that a source tested your exact BERT/GPT/T5 implementation unless it did.
- Add any additional papers that your project guide specifically requires. Keep the literature survey separate from the main research paper if your guide requests separate uploads.